DATA620-W6

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> Here is a dataset that shows a simple 2-node network: the attendance of 18 Southern Women at 14 social events.
>
> Small "musty" datasets like that from this 1941 study have proven very valuable in testing and comparing new network algorithms.
>
> Click here for Python code to create dataset
>
> **What can you infer about the relationships between (1) the women, and (2) the social events?**

Two Mode Networks- Davis Social Southern Club Women

```
import networkx as nx
from networkx.algorithms import bipartite
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
```

Data Load

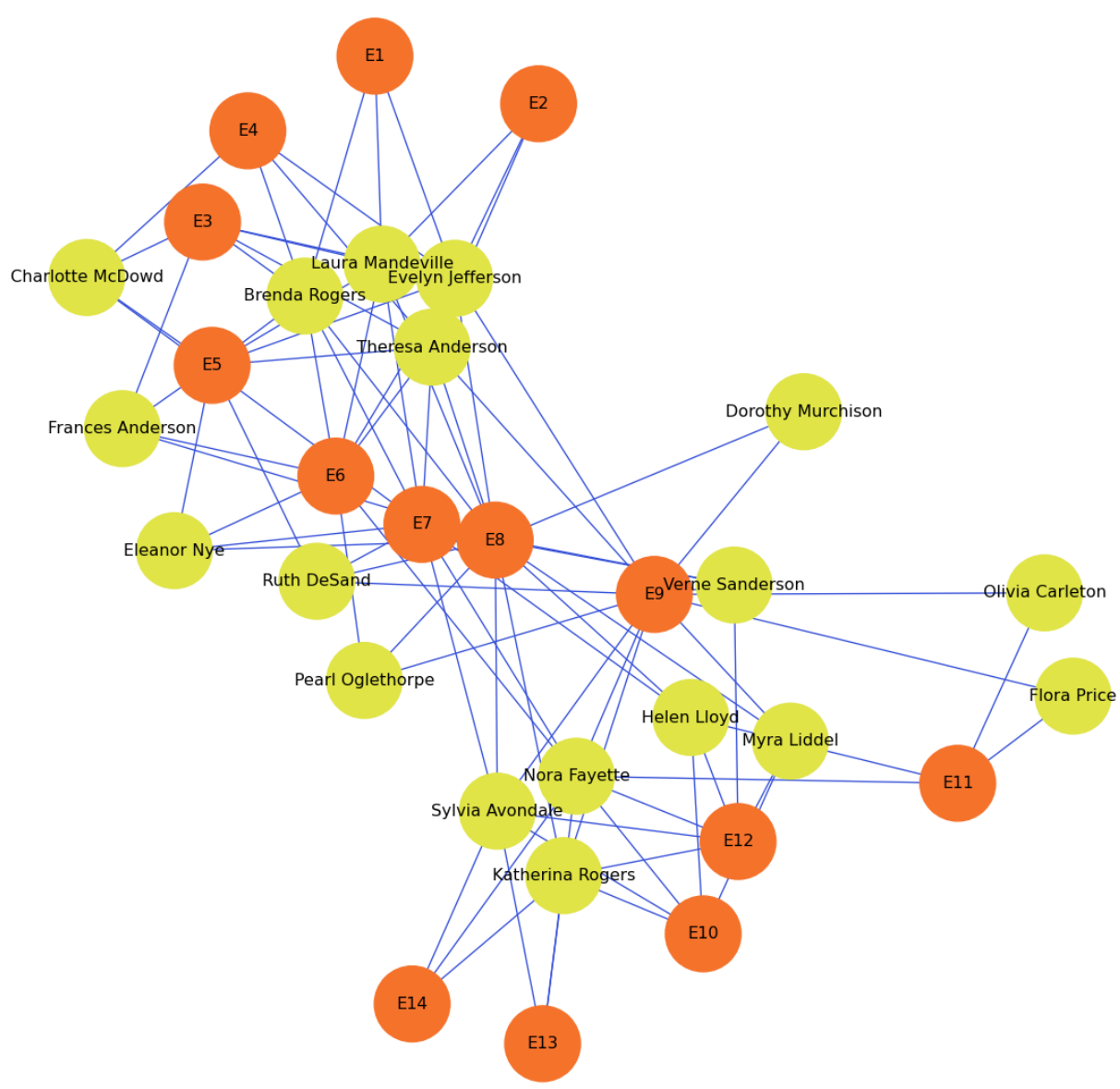
The data can be imported through NetworkX.

```
graph1= nx.davis_southern_women_graph()
```

I'll add some color to the nodes to differentiate between event and person.

```
#--adding some distinction--
nodecolor=["#E0E446" if graph1.nodes[n]["bipartite"]== 0 else "#F5722B" for n in gra

#-- visualizing the graph--
pos= nx.spring_layout(graph1, seed=212)
plt.figure(figsize= (12,12))
nx.draw(graph1, with_labels=True, node_color= nodecolor, node_size=3000, edge_color=
plt.show()
```



While it may be tempting to assume relationships (because of clusters of women in the graph), the the raw graph is really only depicting each woman's relationship to an event, not relationships between the women. To infer about relationships, the graph must be changed into a one-mode network.

First I'll create a matrix. The events should be the columns and the rows will be the women. Attendance will appear as a 1 or a 0. I'll create the data frame from the graph created and and go from there.

```
#-- changing graph edges to df--
edges1= nx.to_pandas_edgelist(graph1)
```

```
#-- changing to df and checking whats what--
nodes1= pd.DataFrame.from_dict(dict(graph1.nodes(data=True)), orient= "index")
nodes1= nodes1.reset_index()
nodes1.columns= ["Node", "Bipartite"]
nodes1
```

	Node	Bipartite
0	Evelyn Jefferson	0
1	Laura Mandeville	0
2	Theresa Anderson	0
3	Brenda Rogers	0
4	Charlotte McDowd	0
5	Frances Anderson	0
6	Eleanor Nye	0
7	Pearl Oglethorpe	0
8	Ruth DeSand	0
9	Verne Sanderson	0
10	Myra Liddel	0
11	Katherina Rogers	0
12	Sylvia Avondale	0
13	Nora Fayette	0
14	Helen Lloyd	0
15	Dorothy Murchison	0
16	Olivia Carleton	0
17	Flora Price	0
18	E1	1
19	E2	1
20	E3	1
21	E4	1
22	E5	1
23	E6	1
24	E7	1
25	E8	1
26	E9	1
27	E10	1
28	E11	1
29	E12	1
30	E13	1
31	E14	1

```
#-- separateing the nodes--
people= nodes1[nodes1["Bipartite"]== 0]["Node"].tolist()
events= nodes1[nodes1["Bipartite"]== 1]["Node"].tolist()

#--creating the matrix--
matrix_a= edges1.pivot_table(index="source", columns="target", aggfunc=len, fill_val
matrix_a
```

target	E1	E10	E11	E12	E13	E14	E2	E3	E4	E5	E6	E7	E8	E9
source														
Brenda Rogers	1	0	0	0	0	0	0	1	1	1	1	1	1	0
Charlotte McDowd	0	0	0	0	0	0	0	1	1	1	0	1	0	0
Dorothy Murchison	0	0	0	0	0	0	0	0	0	0	0	0	1	1
Eleanor Nye	0	0	0	0	0	0	0	0	0	1	1	1	1	0
Evelyn Jefferson	1	0	0	0	0	0	1	1	1	1	1	0	1	1
Flora Price	0	0	1	0	0	0	0	0	0	0	0	0	0	1
Frances Anderson	0	0	0	0	0	0	0	1	0	1	1	0	1	0
Helen Lloyd	0	1	1	1	0	0	0	0	0	0	0	1	1	0
Katherina Rogers	0	1	0	1	1	1	0	0	0	0	0	0	1	1
Laura Mandeville	1	0	0	0	0	0	1	1	0	1	1	1	1	0
Myra Liddel	0	1	0	1	0	0	0	0	0	0	0	0	1	1
Nora Fayette	0	1	1	1	1	1	0	0	0	0	1	1	0	1
Olivia Carleton	0	0	1	0	0	0	0	0	0	0	0	0	0	1
Pearl Oglethorpe	0	0	0	0	0	0	0	0	0	0	1	0	1	1
Ruth DeSand	0	0	0	0	0	0	0	0	0	1	0	1	1	1
Sylvia Avondale	0	1	0	1	1	1	0	0	0	0	0	1	1	1
Theresa Anderson	0	0	0	0	0	0	1	1	1	1	1	1	1	1
Verne Sanderson	0	0	0	1	0	0	0	0	0	0	0	1	1	1

`matrix_a` shows what each woman's attendance is in relation to the events; who attended what.

$$woman \times event$$

From here we'll multiply the matrix by its transpose. The matrix's transpose would each event in relation to who attended.

$$event \times woman$$

So in multiplying matirx_a by its transpose, we get what overlaps, or the women's shared connections. Each value is how many events a pair attended together.

$$woman \times woman$$

```
relationships= matrix_a @ matrix_a.T
relationships
```

source	Brenda Rogers	Charlotte McDowd	Dorothy Murchison	Eleanor Nye	Evelyn Jefferson	Flora Price	Frances Anderson	Helen Lloyd	Katherina Rogers	Laura Mand
source										
Brenda Rogers	7	4	1	4	6	0	4	2	1	6
Charlotte McDowd	4	4	0	2	3	0	2	1	0	3
Dorothy Murchison	1	0	2	1	2	1	1	1	2	1
Eleanor Nye	4	2	1	4	3	0	3	2	1	4
Evelyn Jefferson	6	3	2	3	8	1	4	1	2	6
Flora Price	0	0	1	0	1	2	0	1	1	0
Frances Anderson	4	2	1	3	4	0	4	1	1	4
Helen Lloyd	2	1	1	2	1	1	1	5	3	2
Katherina Rogers	1	0	2	1	2	1	1	3	6	1
Laura Mandeville	6	3	1	4	6	0	4	2	1	7
Myra Liddel	1	0	2	1	2	1	1	3	4	1
Nora Fayette	2	1	1	2	2	2	1	4	5	2

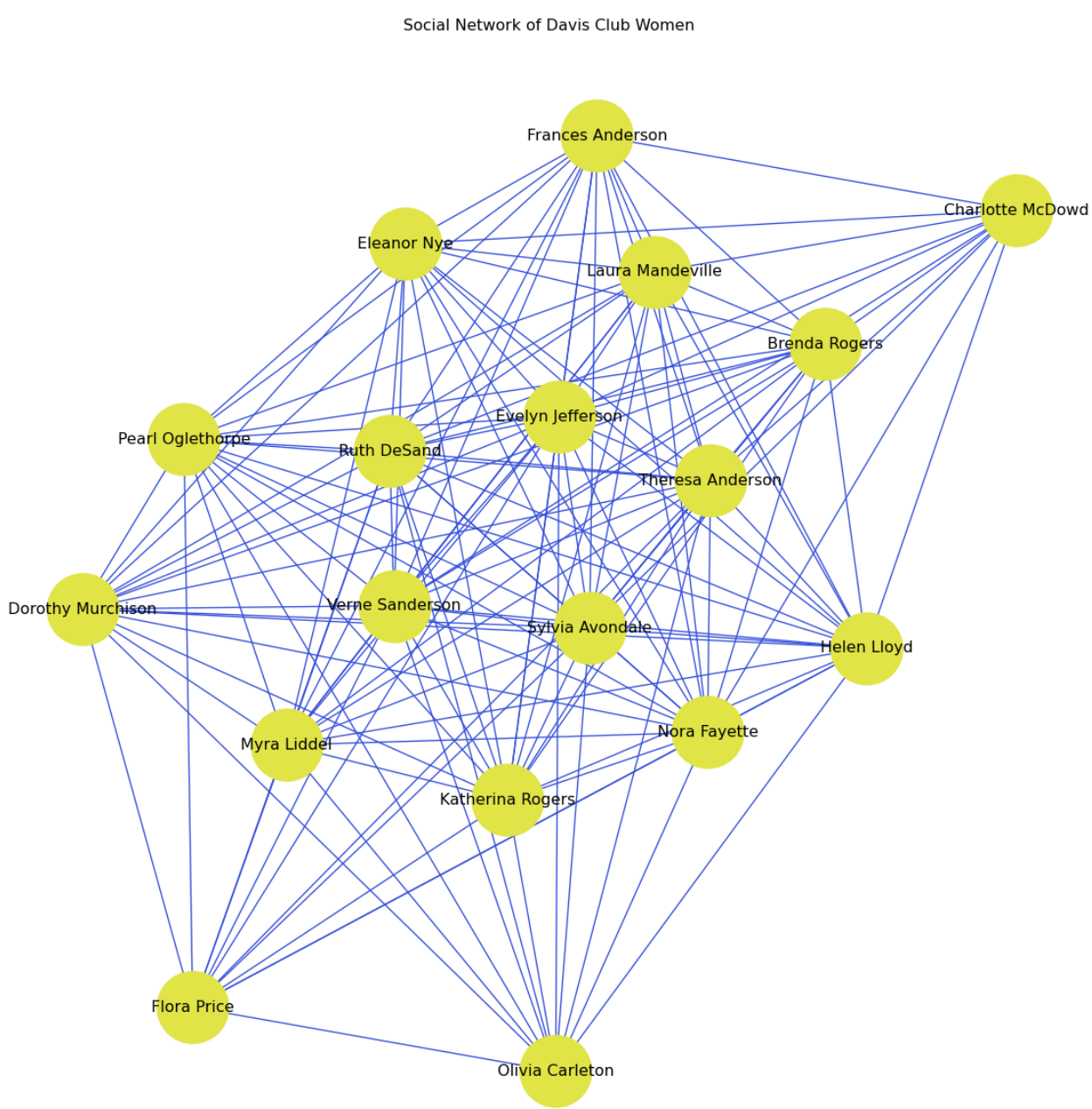
Olivia Carleton	0	0	1	0	1	2	0	1	1	0
Pearl Oglethorpe	2	0	2	2	3	1	2	1	2	2
Ruth DeSand	3	2	2	3	3	1	2	2	2	3
Sylvia Avondale	2	1	2	2	2	1	1	4	6	2
Theresa Anderson	6	4	2	4	7	1	4	2	2	6
Verne Sanderson	2	1	2	2	2	1	1	3	3	2

Now we have the foundation for making assumptions about the social network of the women. I'll remove the women's relationship with themselves (diagonal values) because it'll overpower my visualizations.

I'll visualize the graph:

```
#-- filling the diagonal of matrix--
np.fill_diagonal(relationships.values, 0)
social_club= nx.from_pandas_adjacency(relationships)

#--visualizing--
plt.figure(figsize= (12,12))
nx.draw(social_club, with_labels=True, node_color= "#E0E446", node_size=3000, edge_c
plt.title("Social Network of Davis Club Women")
plt.show()
```



Chapter 5 of Social Network Analysis for Startups (pg 104) defines a function for whats called the **Island Method**. This method basically isolates the connections by weight, creating "islands" of strongly connected networks.

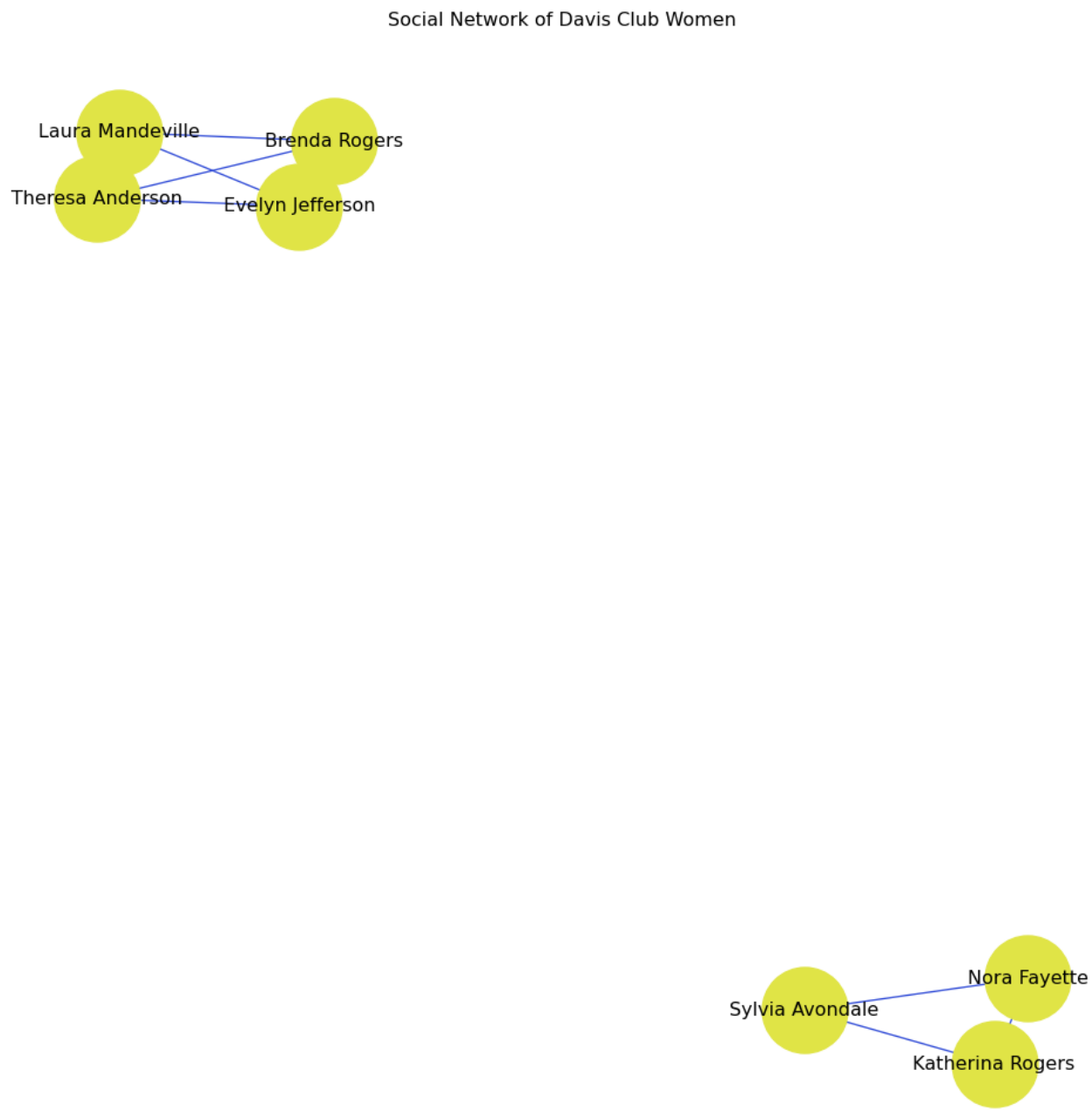
```
#-- chapter 5 function--
def trim_edges(G, weight=1):
    G2= nx.Graph()
    for u, v, d in G.edges(data=True):
        if d["weight"] > weight:
            G2.add_edge(u, v, **d)
    return G2
```

The distribution of the treshold/weights will help determining what number to select for the trimming function.

```
#--determining threshold to use in func--
weights= list(nx.get_edge_attributes(social_club, "weight").values())
pd.Series(weights).value_counts().sort_index()

#-- running the function--
trimmed= trim_edges(social_club, weight= 4)

#--visualizing--
plt.figure(figsize= (10,10))
nx.draw(trimmed, with_labels=True, node_size=3000, node_color= "#E0E446", edge_color
plt.title("Social Network of Davis Club Women")
plt.show()
```



Most edges are gathered at threshold 1 and 2, selecting 3 or 4 as the threshold would be best.

The new graph shows the most central social circles within the Davis club.

```
social_edges= nx.to_pandas_edgelist(social_club).sort_values("weight", ascending=Fal
print(social_edges.head(5))
```

```
               source            target  weight
63   Evelyn Jefferson  Theresa Anderson       7
7       Brenda Rogers  Laura Mandeville       6
3       Brenda Rogers  Evelyn Jefferson       6
101  Katherina Rogers   Sylvia Avondale       6
56   Evelyn Jefferson  Laura Mandeville       6
```

Evelyn and Brenda appear twice as sources. We can infer Evelyn, Brenda, and Katherina are central and important in this social club. The islands depicted Evelyn and Brenda are in one social circle, and Katerina is central to another island that includes Sylvia and Nora.

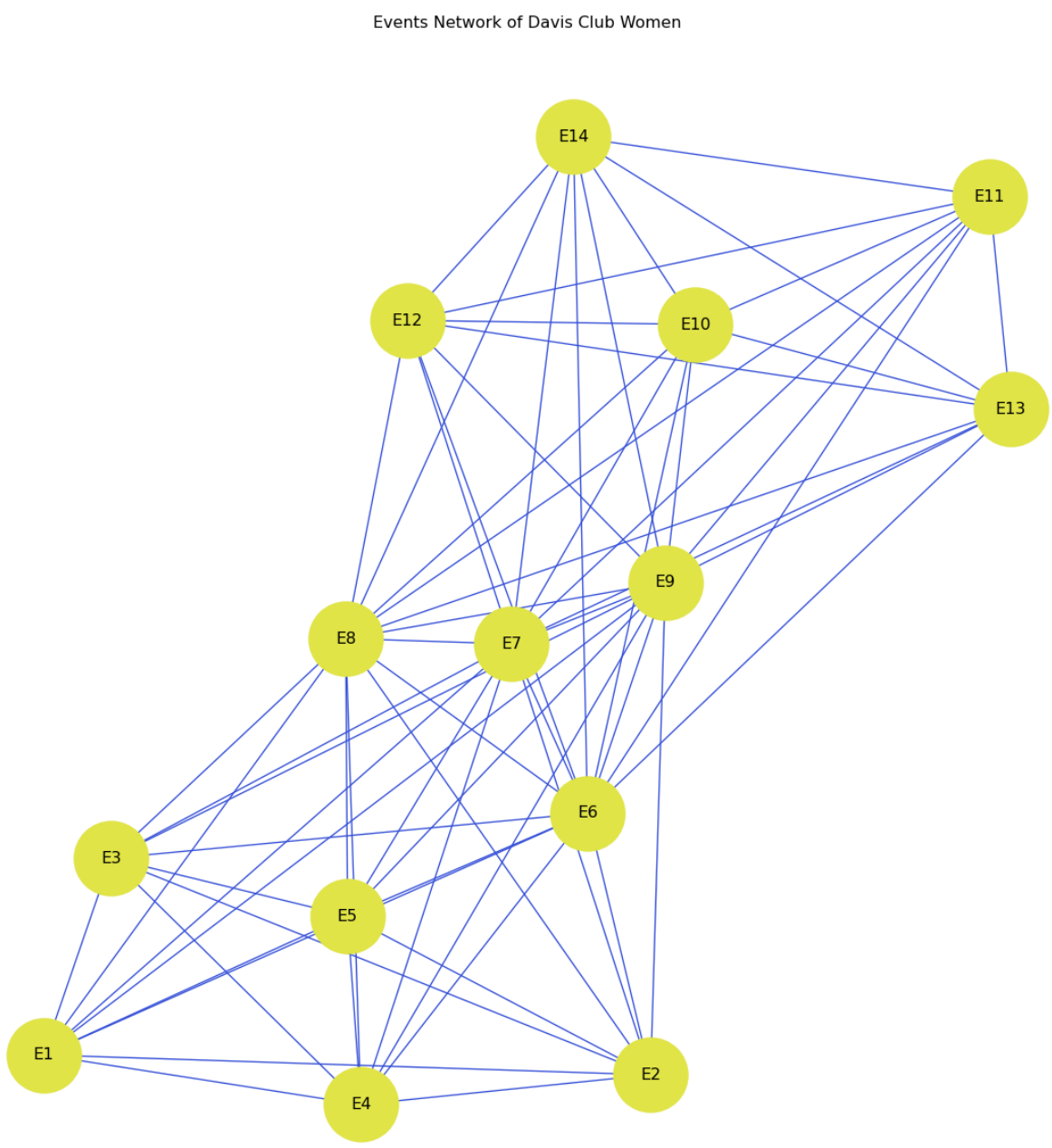
Event Networks

To make inferences about the events, we'll flip the matrix * matrix transpose and quickly redo the same analysis.

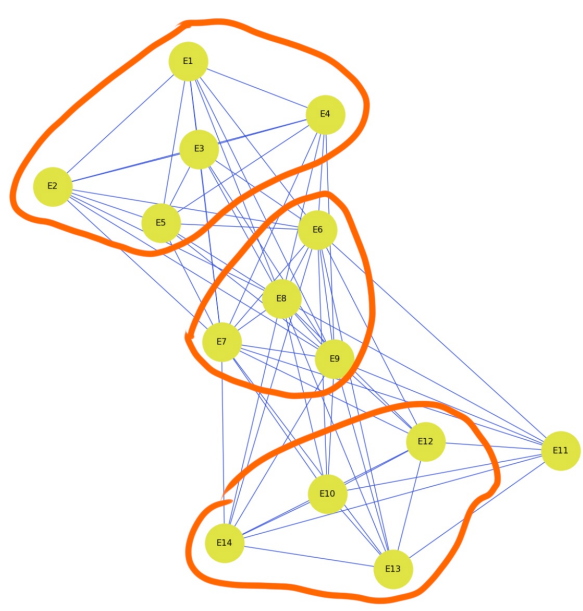
```
#-- overlap matrix--
event_rel= matrix_a.T @ matrix_a
```

```
#--removing the diag values--
np.fill_diagonal(event_rel.values, 0)
event_g= nx.from_pandas_adjacency(event_rel)

#--visualizing--
plt.figure(figsize= (12,12))
nx.draw(event_g, with_labels=True, node_color= "#E0E446", node_size=3000, edge_color
plt.title("Events Network of Davis Club Women")
plt.show()
```



The network has some slight clustering:



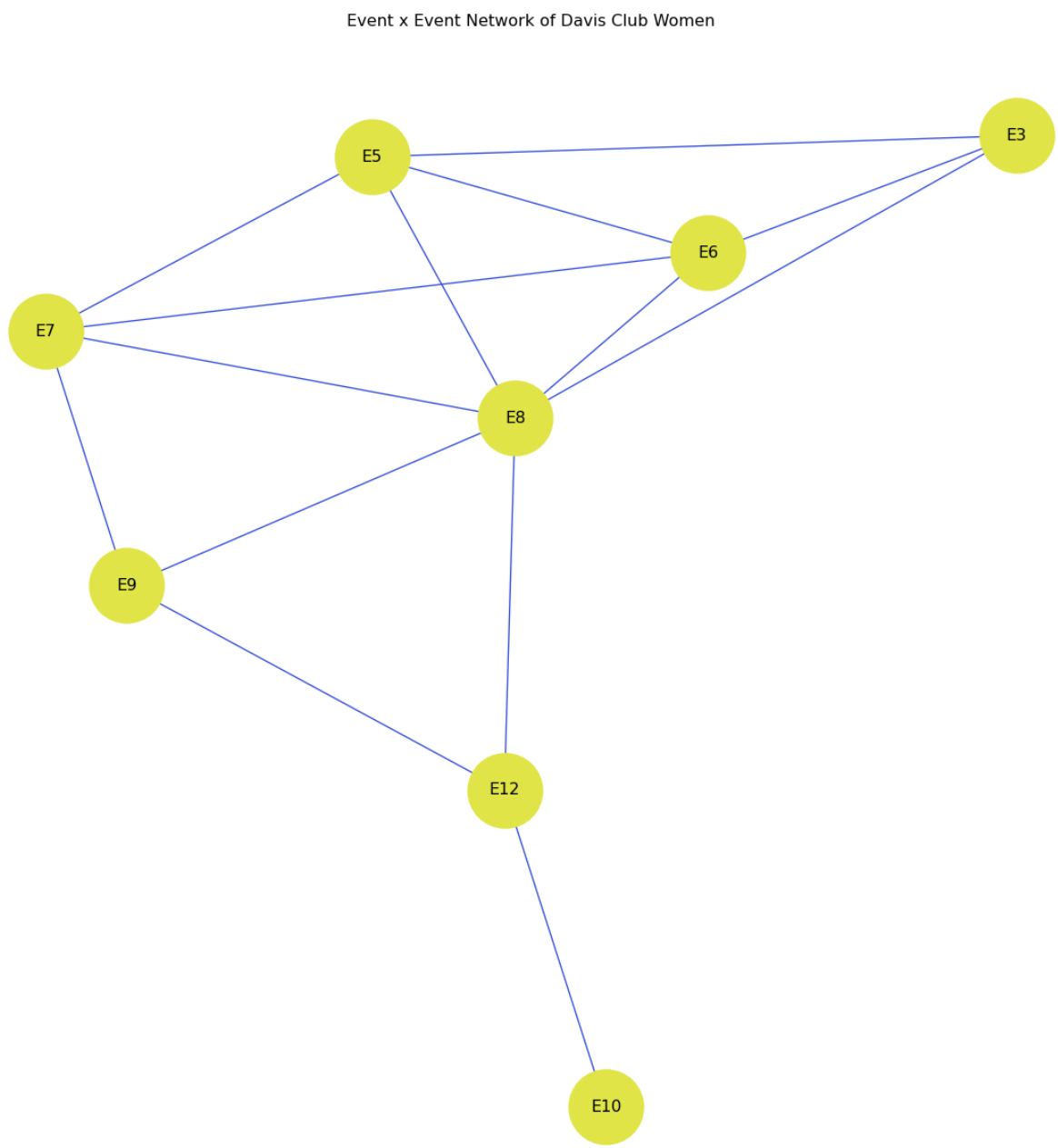
We'll continue by trimming and using the output to infer:

```
#--picking a threshold--
weights2= list(nx.get_edge_attributes(event_g, "weight").values())
pd.Series(weights2).value_counts().sort_index()

#-- trim function--
trimmed2= trim_edges(event_g, weight= 4)

#--visualizing--
plt.figure(figsize= (12,12))
nx.draw(trimmed2, with_labels=True, node_color= "#E0E446", node_size=3000, edge_colo
plt.title("Event x Event Network of Davis Club Women")
plt.show()
```

```
#--List
event_edges= nx.to_pandas_edgelist(event_g).sort_values("weight", ascending=False)
print(event_edges.head(5))
```




```
   source target  weight
65     E8     E9       9
63     E7     E8       8
61     E6     E8       7
58     E5     E8       7
56     E5     E6       6
```

E8 appears 4 times and is identified as central, connecting directly to E5, E6, E7, and E9.

Conclusion

In modifying a two mode network, into a one mode network, one can gain accurate insights into social networks' relationship structure. The island method helps identify relationships that are most central by implementing a threshold. Within the Davis Club member relationship structure, we can infer that some of the women have these *island* connections. I think Brenda, Evelyn, and Katherina are central to inner social circles within the club. We can also infer that Event 8 was a hub within the events network, many of of the same members attended along with the events E8 was tied to (6,7,8 and 9).

Bibliography:

Tsvetovat, M., & Kouznetsov, A. (2011). Social Network Analysis for Startups: Finding connections on the social web (1st ed.). O'Reilly Media.

Social Network Analysis: Analyzing Two-Mode Networks:

https://www.youtube.com/watch?v=o0Mxjll7-6o